Photonic Interconnects for TPU LLM Workloads

Pranav Sitaraman
sitaraman.pranav@gmail.com

Simone Surdi
simonesurdi@gmail.com

AccelForge with TCM (arXiv:2602.15172) finds optimal per-chip mappings, but it has no concept of the network between chips. We added one. The network model prices each transfer by hop count, contention, multicast support, and a per-bit energy split into a conversion term and a per-mm transmission term. An iterative loop pushes that cost into a logical cluster-memory level so that when we change the fabric the mapper retunes against network-augmented memory. Above the mapper sits a discrete distributed-schedule layer with four XLA/GSPMD-style policies, and on top of that a stage model for a 9216-chip Ironwood pod. We compare Lightmatter Passage (EVK50, EVK100, L200, L200X, M1000) against an 800G electrical baseline and a strong 3.5 Tb/s, 6 pJ/bit chiplet baseline on a 16-node cluster, a 64-chip $4\times4\times4$ TPU cube, and the 9216-chip superpod, with six LLM-stage GEMM proxies plus Gemma 4 and Gemini-class workloads.

I. Key Insights and Innovations

1) **Photonics doesn't move the inner mapping. It moves the schedule.** Across 24 (workload, technology, scenario) combinations on the 64-chip cube, AccelForge picked the same partitioning, dataflow, and tile shape every time — 0/24 inner mappings changed. The wins live one layer up. The optimal inter-chip communication pattern flipped on 5 of 16 workload-scenario pairs, and on KV attention switching EVK100 from sharded collectives to remote pooled-HBM streaming cut compute energy by 73.86%. Treating the per-chip mapping and the inter-chip schedule as one number hides this, so we report them separately.
2) **The advantage isn't bandwidth but instead multicast and distance-insensitive transport.** We picked the chiplet baseline (3.5 Tb/s, 6 pJ/bit) specifically to rule out the "photonics-is-just-more-bandwidth" possibility. M1000 still beats it by 99.85% on shared-DRAM multicast, because N unicasts collapse to one broadcast and contention drops to zero on a switched fabric. Our link model separates conversion from transmission, so electrical pJ/bit grows with distance while photonic pJ/bit stays flat. We find that per-chip multicast cost is generally flat from 64 to 9216 chips for M1000 while torus and chiplet fabrics face higher costs as N grows.
3) **Photonics doesn't help everything equally.** On the 9216-chip Ironwood superpod, the M1000 stack delivers 42.43× inference speedup and 94.05% energy reduction on the Gemma 4 26B MoE proxy workload equivalent, while the long-context Gemini Deep Think proxy gets 14.50×. Training is much smaller, 3.37× and 1.43× respectively. Once collectives shrink, training hits an Amdahl ceiling at $\max(T_{\text{compute}}, T_{\text{HBM}})$ that photonics doesn't move; inference stays exposed because decode and KV/expert traffic cross the fabric every step. The strong case is inference and long-context serving, not training.

II. Contributions

Simone Surdi. Network model and AccelForge plumbing. Wrote the topology classes (ring, mesh, 3-D torus, optical crossbar / switched fabric) and the `LogicalClusterMemory` level. Wrote `solve_iteratively`, which re-runs the mapper with updated logical-memory energies and latencies until they stop moving. Built the mapping-signature comparison and ran the cluster, TPU-cube, and chiplet-baseline studies. Pulled and verified the technology constants against Lightmatter (SC25, ISES, product pages) and TPU (Cloud docs, Ironwood blog, Hot Chips). Ran the bandwidth and energy sensitivity sweeps. Helped write the report, poster, and technical summary.

Pranav Sitaraman. Distributed-schedule and superpod analysis. Wrote the distance-parametrized link model and the OCS/OPS sensitivity sweep. Implemented the four distributed schedules (SPMD collective, remote pooled-HBM streaming, replicated cache, photonic multicast cache fill). Built the 9216-chip Ironwood stage model (prefill, decode, refill, MoE, training, gradient, optimizer) with the α_{ovl} overlap factor. Ran strong and weak scaling against the Gemma 4 anchors and Gemini-class proxies. Built the ablation ladder and the 71-test integration suite. Helped write the report, poster, and technical summary.

Joint. Designed the four-layer pipeline (workload → architecture → AccelForge mapping → distributed schedule), worked through the experimental setup.

III. Artifact

- **GitHub Repository:** https://github.com/PranavSitaraman/photonic-interconnects
- **Report PDF:** https://drive.google.com/file/d/1YP_ElsrpxuoX1Yr9KAZqQFVg7-6iGiRp/view?usp=sharing
- **Poster PDF:** https://drive.google.com/file/d/1zd9CNyCml5L4fFaEEHRt–8OAQqAhu6B/view?usp=sharing